

3.3 ECONOMIES & DISECONOMIES OF SCALE

Multiple Choice Questions

1990/CE/II/22

'As the carrying capacity of a bus increases, the oil consumption increases. However, the oil consumption increases at a slower rate than the carrying capacity. This is an example of

- A. external economies of scale
- B. technical economies of scale
- C. management economies of scale
- D. research economies of scale

1991/CE/II/20

The cost per tonne of goods transported by a bigger lorry is lower than that of a smaller one. This is an example of

- A. management economies
- B. marketing economies
- C. financial economies
- D. technical economies

1992/CE/II/20

The Urban Council and Regional Council are responsible for growing young trees for planting around Hong Kong. Each nursery is usually large in size. Furthermore, a large number of the same kind of young trees are planted in the same area of the nursery. This type of nursery has the following advantages:

- (1) economy of scale on irrigation facilities
- (2) economy of scale by caring for plants of the same kind
- (3) economy of scale in purchasing fertilizers

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

1993/CE/II/21

'It is uneconomical to use a jumbo aircraft to carry a small number of passengers.' This can be explained by

- A. the principle of comparative advantage
- B. the law of diminishing returns
- C. technical economies
- D. diseconomies of scale

1995/CE/II/22

Which of the following about economies of scale is true?

- A. It applies to short-run as well as long-run production.
- B. It will lead to a decrease in the average cost of production when the output increases.
- C. It occurs only when scale of production is sufficiently large.
- D. It will lead to a decrease in the total cost of production when the output increases.

1998/CE/II/13

In general, which of the following are the advantages of a small firm over a large firm in the same industry?

- (1) There is a better relationship between the employees and the employer.
 - (2) It is more adaptable to changing market situations.
 - (3) Specialization can be carried out more extensively.
 - (4) More funds can be raised at a lower cost.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (4) only
 - D. (3) and (4) only

2000/CE/II/17

Which of the following implies the existence of economies of scale?

- A. The total cost of production decreases when output increases.
- B. The average cost of production decreases when the production scale increases.
- C. Profit increases when output increases.
- D. The amount of output increases by a smaller proportion than that of inputs.

2004/CE/II/20

Compared with small stores, large supermarkets purchase a large stock of goods from suppliers at a lower average cost. It shows that these large supermarkets enjoy _____ economies of scale.

- A. finance
- B. management
- C. technical
- D. marketing

2005/CE/II/18

When a firm enjoys economies of scale, it is in the _____ and its _____ is decreasing.

- A. short run average cost
- B. long run average cost
- C. short run total cost
- D. long run total cost

2006/CE/II/18

If large publishers are more able to fully utilize their machinery to lower the average cost than small publishers, then these large publishers enjoy _____ economies of scale.

- A. financial
- B. management
- C. marketing
- D. technical

2008/CE/II/17

Compared with local small accounting firms, large accounting firms in Hong Kong are more willing to use resources to familiarize themselves with China's laws in order to expand their business into Mainland China. One possible reason is that large accounting firms enjoy

- A. financial economies of scale.
- B. technical economies of scale.
- C. marketing economies of scale.
- D. economies of scale in research and development.

2013/DSE/I/05

In order to increase the number of nurses for expanding medical services, the government has allocated more resources to expand the nursing programme in universities. In this case, private hospitals will

- A. enjoy internal economies of scale.
- B. suffer internal diseconomies of scale.
- C. enjoy external economies of scale.
- D. suffer external diseconomies of scale.

2016/DSE/I/07

A school is a production unit which provides educational services. The following are people and objects that are usually found at a school:

- (1) the school building
- (2) teachers
- (3) students
- (4) private cars owned by teachers
- (5) sanitisers (洗手液) in the toilets

In the long run, which of the above items are variable factors of production of the school?

- A. (1) and (5) only
- B. (1), (2) and (5) only
- C. (2), (3) and (4) only
- D. (1), (2), (3), (4) and (5)

2018/DSE/I/12

Which of the following statements about economies of scale is correct?

- A. It could be a source of monopoly power.
- B. It exists when the average fixed cost is falling.
- C. Only large firms enjoy economies of scale.
- D. It is the result of the law of diminishing marginal returns.

2020/DSE/I/7

Tai On Building (太安樓) in eastern Hong Kong Island is well-known as a 'foodie paradise'. A variety of street snacks such as cart noodles, egg waffles, dumplings and mango mochi are sold by numerous small vendors. Customers from different districts fill their stomachs at Tai On Building.

Based on the above information, street snack vendors at Tai On Building can_____.

- A. enjoy internal economies of scale
- B. enjoy external economies of scale
- C. suffer internal diseconomies of scale
- D. suffer external diseconomies of scale

2022/DSE/I/07

The following table shows the input, output and cost relationships of a firm. The only inputs required for production are labour and capital.

Labour (units)	Capital (units)	Total product (units)	Total cost of production
1	1	20	\$150
2	1	50	\$200
3	1	70	\$250
4	1	80	\$300

Based on the above table, we can conclude that

- A. the firm enjoys economies of scale.
- B. the optimal scale of production is 70 units.
- C. the average product of labour is at the highest when the quantity of labour is 2 units.
- D. the law of diminishing marginal returns does not apply to the firm.

2025/DSE/I/11

Miss Chan plans to open a flower shop. She is considering whether to open a flower shop at Flower Market Road, a district famous for having lots of flower shops, or to open at another location with no flower shops nearby.

Based on the above information, an advantage for Miss Chan to open a flower shop at Flower Market Road is that she may be able to _____.

- (1) enjoy internal economies of scale in terms of lower transportation costs
 - (2) enjoy external economies of scale in terms of lower marketing costs
 - (3) sell flowers at a higher price due to keener competition among sellers
- A. (1) only
 - B. (2) only
 - C. (1) and (3) only
 - D. (2) and (3) only

Short & Structured Questions

1992/CE/I/1(b)

Suggest **TWO** reasons why it can be more advantageous to run a large taxi company with many taxis rather than a small taxi company with only one taxi. (6 marks)

1993/CE/I/1(c)

Yummy Yummy is a fast food shop.

(i) Last year, 'Yummy Yummy' planned to establish a branch. Explain whether this was a short-run or long-run plan. (2 marks)

(ii) After the opening of this branch, the unit cost of producing hamburgers is lower. Give **TWO** economic reasons for this. (6 marks)

1995/CE/I/10(a)(ii)

Study the following piece of news reported in the USA:

A local singer has volunteered to perform two identical shows and the sales receipt of the concert tickets will be donated to the Rwanda refugees in Africa for their food and medical supply expenses

Explain **TWO** economies of scale that may arise from holding two identical shows instead of only one show. (4 marks)

1997/CE/I/9(a)

Suppose the government builds a railway in northwest New Territories.

If the new railway is to be run by the Kowloon-Canton Railway Corporation (KCRC), give **TWO** reasons why this can bring down the average cost of KCRC. (4 marks)

2000/CE/I/10(b)(i)

Suppose Bank X of economy A takes over Bank Y of economy B by buying up its controlling share worth \$5 million. Explain what type of expansion the takeover belongs to and explain **THREE** economies of scale that may arise from the takeover. (8 marks)

2004/CE/I/11(c)(i)

A famous chain of fashion stores opens a new branch.

Explain **TWO** economies of scale that the chain stores might enjoy due to the expansion. (4 marks)

2007/CE/I/1

It is expected that the average cost of providing the railway service will decrease after the integration of the Mass Transit Railway Corporation Limited (MTRC) and the Kowloon-Canton Railway Corporation (KCRC).

Give **THREE** reasons to explain why such integration could reduce the average cost. (6 marks)

2009/CE/I/9(a)

In recent years, the price of textbooks has become a public concern.

Textbook publishers usually print textbooks in large amounts in order to reduce their average cost of production. Explain **TWO** economies of scale that a firm may enjoy when it expands its production scale. (4 marks)

2010/CE/I/9(a)

Ocean Park Hong Kong is undergoing a major redevelopment plan.

Ocean Park will expand to double the number of its attractions. Explain **TWO** economies of scale that it may enjoy after expansion. (4 marks)

2015/DSE/II/2(b)

Many electronic product retailers are located in Mong Kok.

Suggest **TWO** reasons why an electronic product retailer in Mong Kok may enjoy external economies of scale. (4 marks)

2016/DSE/II/2(c)

The average cost of providing the services decreases when the Group operates more tutorial classes in different districts. Give

THREE reasons to explain why its average cost decreases. (3 marks)

2017/DSE/II/2(b)

Due to the opening of several new branches, the average cost of production of the restaurant has increased. Explain with **TWO** reasons why this happens. (4 marks)

2018/DSE/II/3

Shenzhen has become China's 'Silicon Valley'. The headquarters of many high technology firms such as Tencent (騰訊), Huawei (華為) and BYD (比亞迪) are located there. Explain **TWO** external economies of scale high technology firms may enjoy in Shenzhen. (4 marks)

2020/DSE/II/2

A chain restaurant providing Chinese cuisine set up a cake shop.

- (c) After the opening of 10 more restaurants, the firm found that its average cost decreased. State **TWO** possible reasons for this. (2 marks)

2022/DSE/II/1(b)

- (b) After opening some new restaurants, Mary found that the average cost of production of her company had increased. Explain the change in average cost with **ONE** reason. (2 marks)

2023/DSE/II/11(d)

In Hong Kong, fans of popular singers often complain that they are unable to buy concert tickets of their idols at the official prices.

- (b) Which function of price is illustrated in the above situation? Explain your answer. (2 marks)

2024/DSE/II/11(f)

Ann is the owner of a coffee shop in Hong Kong.

- (f) After opening several new coffee shops, Ann found that the average cost of production of her business had decreased. Give **TWO** reasons for the above change in average cost. (2 marks)